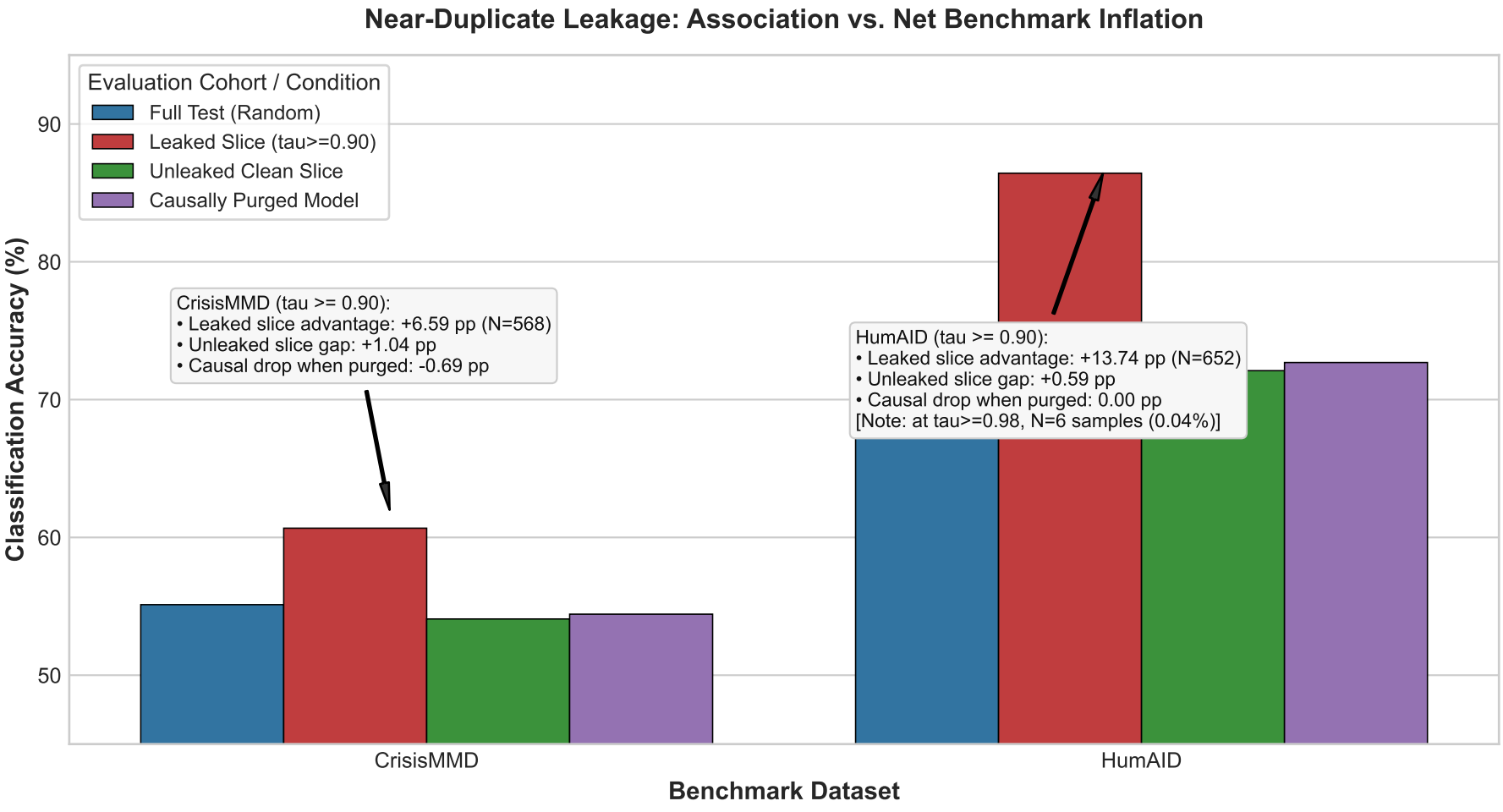
Near-Duplicate Leakage: Association vs. Net Benchmark Inflation
Evaluation Cohort / Condition
Full Test (Random)
Leaked Slice (tau>=0.90)
Unleaked Clean Slice
Causally Purged Model
CrisisMMD (tau >= 0.90):
• Leaked slice advantage: +6.59 pp (N=568)
• Unleaked slice gap: +1.04 pp
• Causal drop when purged: -0.69 pp
HumAID (tau >= 0.90):
• Leaked slice advantage: +13.74 pp (N=652)
• Unleaked slice gap: +0.59 pp
• Causal drop when purged: 0.00 pp
[Note: at tau>=0.98, N=6 samples (0.04%)]
Classification Accuracy (%)
90
80
70
60
50
CrisisMMD
HumAID
Benchmark Dataset